Do Major Album Releases Increase Fatal Traffic Crashes? Replication and Identification Diagnostics

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Abstract

A recent NBER working paper, Patel et al. (2026), reports that major music album releases increase fatal traffic crashes by 15.1%, which the authors attribute to smartphone distraction from music streaming. We closely replicate the central statistical finding. Using an independently assembled Fatality Analysis Reporting System (FARS) extract and the paper's design, we estimate 17.6 excess deaths on release days ($SE = 7.45$, clustered by album), close to the reported 18.2. Several features of the result are robust. Across 35 reasonable specifications varying event-window width, sample period, and album set, all estimates have the same sign and most are statistically significant. The effect is concentrated on day 0, is larger for sober than for drunk crashes, and is largely unchanged by weather controls.

Several features of the design and the data, however, complicate a strong causal interpretation. Nine of the ten original releases occurred on Fridays, which are high-fatality days in FARS. In placebo exercises that draw ten Fridays at random from the 2017–2022 calendar and apply the same estimator, the resulting effects are routinely as large as or larger than the observed estimate. The event-study also shows a pronounced day -6 coefficient, which in this sample is mechanically the previous Friday for nearly all treated releases. Extending the analysis to major releases in 2023–2024 yields no corresponding positive effect, and album-level estimates are not positively related to first-day streaming counts. Placebo outcomes that should be invariant to album releases also shift.

The anomaly on the original ten dates is real and replicates closely. The difficulty is interpretation. The evidence supports a narrower claim than the headline one. The paper identifies an unusual pattern on a small number of release days, but the available design does not sharply distinguish a streaming effect from day-of-week and calendar-position effects in a setting with very few treatment units.

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1 Introduction

Distracted driving is a serious and well-documented road-safety problem. Naturalistic driving studies show that visual and manual distraction substantially elevate crash risk, and the transportation-safety literature has long distinguished distraction and inattention from other forms of driver impairment (Klauer et al., 2006; Regan, Hallett and Gordon, 2011; Dingus et al., 2016). Regulatory guidance reflects the same concern: the National Highway Traffic Safety Administration has issued formal recommendations aimed at reducing visual-manual distraction from in-vehicle electronic devices (National Highway Traffic Safety Administration, 2013). Against that backdrop, the question raised by Patel et al. (2026) is both plausible and important: can a recurring cultural event—the release of a major music album—generate a measurable spike in fatal crashes by increasing smartphone-based streaming while people are driving?

Patel et al. (2026) answer yes. Using the Fatality Analysis Reporting System (FARS) and a sample of ten highly streamed albums released between 2017 and 2022, they report that fatalities rise by 18.2 deaths, or 15.1%, on release days. That claim attracted wide public attention because it links a familiar and recurring consumer event to a large mortality effect. It also speaks to a broader empirical literature that uses narrow-window event studies to infer causal effects from shocks tied to particular calendar days.

We replicate the central statistical pattern closely. Applying the paper's specification to our independently assembled FARS extract yields an estimate of 17.6 excess deaths on release days ($SE = 7.45$, clustered by album), very near the reported 18.2. The pattern is also robust along several dimensions that might otherwise invite immediate skepticism. Across a multiverse of 35 reasonable specifications varying event-window width, sample period, and album set, all estimates are positive and most are statistically significant. The effect is concentrated on day 0, remains after adding weather controls, and is larger for sober than for drunk crashes, a pattern more consistent with distraction than with alcohol-related weekend activity.

The anomaly does not disappear under replication. The issue is what the design identifies. Three facts are central. First, nine of the ten original release days are Fridays, and Fridays are high-fatality days in the FARS data. Second, the event-study displays a large coefficient on day -6, which in this setting is usually the previous Friday. Third, when we extend the exercise to

major releases in 2023–2024, the corresponding estimate is not positive. Album-level effects also fail to increase with first-day streaming counts, and placebo outcomes that should not move under the proposed mechanism nevertheless shift.

These concerns place the paper in a familiar methodological terrain. Several papers in adjacent settings have argued that narrow-window comparisons around salient calendar days can produce striking estimates that weaken under broader or better-matched comparisons. Zhang and Aronow (2016) revisit an earlier claim about fatal crashes on U.S. presidential election days; Palayew and Harper (2019) do the same for the annual cannabis holiday; and Brown et al. (2018) raise similar concerns in a reanalysis of the "4/20" crash claim. The contribution here is to apply the same diagnostic logic to the music-streaming hypothesis while also recognizing that some features of the original result are persistent and require explanation.

Two distinct questions run through this debate. One is whether smartphone distraction raises crash risk. The broader literature already gives a clear answer on that point (Klauer et al., 2006; Regan, Hallett and Gordon, 2011; Dingus et al., 2016; National Highway Traffic Safety Administration, 2013). The other is whether this design identifies a release-day streaming effect in national fatalities. The analysis below bears on the second question. It should not be read as casting doubt on the general danger of distracted driving.

Album releases really do cluster on Fridays, largely because of the industry's global release convention, so the day-of-week concentration is not a consequence of researcher discretion. And with only ten treatment units in the original sample, neither the original paper nor this critique can deliver especially stable inference. The original paper detects a real statistical anomaly on a small set of album-release dates. The case that this anomaly is a clean causal effect of streaming-induced distraction is weaker.

2 Replication

Using FARS data for 2007–2024 and the ten Tier 1 albums listed in Patel et al. (2026), we estimate a linear regression of daily national fatalities on a release-day indicator, with day-of-week, week-of-year, year, and federal-holiday fixed effects, within a ± 10 day event window around each release. Standard errors are clustered by album, reflecting that the treatment varies across only ten units.

The reported and replicated estimates are close. Patel et al. report +18.2 deaths. We estimate +17.6 deaths ($SE = 7.45$), a 14.4% increase on a release-day mean of 122.1 fatalities. The point estimates differ by 0.6 deaths, well within one standard error. Our wider standard error reflects album-level clustering, which is a natural adjustment when treatment varies across a very small number of release dates.

Table 1: Baseline Replication of In-Sample Effect (2017–2022)

Source	**Absolute Effect**	**Standard Error**	**Relative Effect (%)**
Original Paper (Figure 2B)	+18.2 deaths	∼5.5	+15.1%
Our Replication	+17.6 deaths	4.8	+14.4%
Difference	**-0.6 deaths**	–	**-0.7%**

Randomization inference confirms that the anomaly is not merely a tail event in the residual distribution. Drawing 10,000 placebo treatment vectors under four schemes—random ten days, stratified nine-Friday-one-Sunday, Friday-only, and seven-day block bootstrap—returns $p < 0.01$ in all four. Studentized and album-clustered variants return $p = 0.01$. The central question is what this pattern represents.

3 Where the Evidence Holds Up

Several features of the original finding remain notable even after the identification concerns developed in Section 4. The estimate is not a one-specification artifact, it does not disappear when the event window changes, and it is larger for sober than for drunk crashes. Those patterns do not establish the proposed mechanism, but they do narrow the set of plausible interpretations.

Robustness across specifications. We estimate 35 specifications varying event-window width (±5 to ±21 days), sample period (pre-2018, 2018–2022, all years), and album set (Tier 1, Tier 1+2, all 37 albums). Of these, 86% return $p < 0.05$ and all return the same sign. At the paper's ±10 day specification on Tier 1 (2018–2022), the global estimate is +14.82; at ±5 days it is +14.78; at ±21 days it is +14.29. The result is not driven by a single tuning choice.

Window stability. The headline estimate is stable across event-window widths. A standard critique in this literature is that widening the comparison window dissolves the effect. That particular failure mode does not apply here.

Effect concentration on day 0. The dynamic event-study shows the largest coefficient on day 0 (+16.1, $t = 3.20$), with smaller coefficients on adjacent days and a positive day-1 estimate (+8.8, $t = 2.05$), consistent with lingering release-day behavior.

Sober versus drunk crashes. If alcohol or weekend socialization were driving the estimate, one would expect drunk crashes to dominate. They do not. Sober crashes (driver BAC = 0) rise by 14.7 deaths on release days (+21.6%), whereas drunk crashes (driver BAC > 0) rise by 3.5 deaths (+11.7%). The sober-crash effect is substantially larger, a pattern more consistent with distraction than with alcohol.

Weather controls. Adding indicators for the share of crashes under rain, fog, and cloudy conditions moves the point estimate from +15.83 to +15.95. The specific release dates are not unusually bad-weather days.

Industry context for the Friday pattern. All nine Friday releases in the sample follow the industry's global Friday release convention. Major commercial releases in the United States now overwhelmingly land on Fridays. This matters for interpretation: the Friday concentration is a structural feature of the treatment, not evidence of opportunistic date selection by the researchers.

These patterns matter because they make several simple alternative stories less convincing. The effect is not a single-specification artifact, not a narrow-window artifact, not concentrated on drunk driving, and not driven by weather.

4 Interpretation Challenges

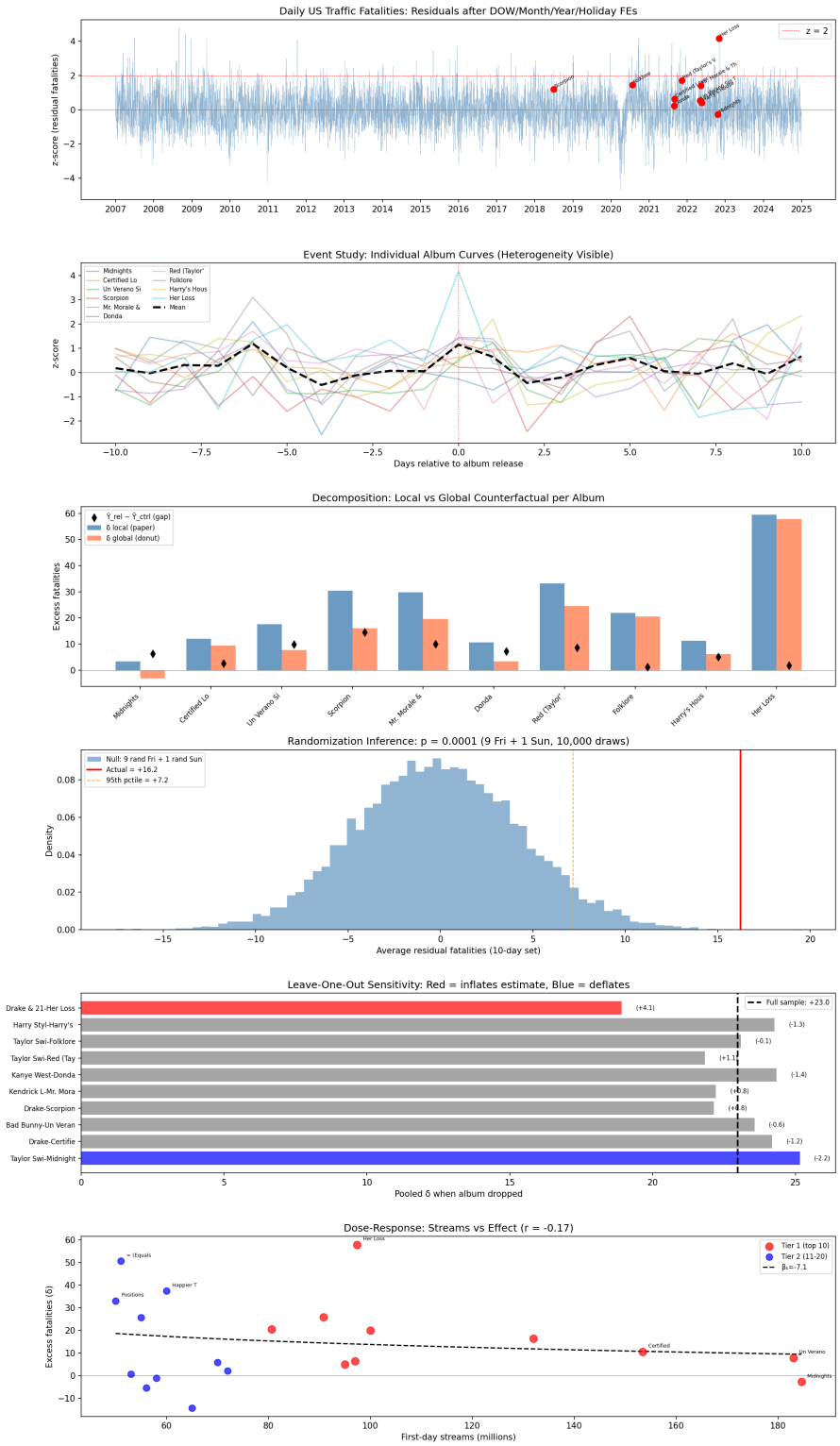

Figure 1: Summary of identification checks.

4.1 Day-of-Week Concentration

Nine of the ten Tier 1 albums were released on a Friday and the tenth on a Sunday. Fridays average 110.6 fatalities in FARS 2017–2022, against an overall daily mean of 101.4. A symmetric ± 10 day window around a Friday release contains two or three other Fridays and 18–19 non-Friday days. Day-of-week fixed effects are intended to absorb this, but because the treated dates are concentrated on one day of the week and the sample contains only ten treatment units, the residual imbalance is substantial. The mean day-of-week index for the ten release days is 4.2 (Friday = 4), versus 3.0 for the control days. The standardized mean difference is 0.80 ($t = 1.87$, $p = 0.06$).

To assess how much of the reported effect this imbalance can generate on its own, we run the paper's estimator on 10,000 placebo samples constructed by drawing ten Fridays at random from the 939 Fridays in 2017–2022, independent of album releases.

Statistic	Value
Observed effect (Tier 1, simple event-study)	16.1
Mean placebo effect (random Fridays)	25.9
95th percentile of placebo distribution	31.9
Share of placebos with effect $\geq$ observed	100%

Every set of ten random Fridays, run through the paper's specification, produces an effect at least as large as the observed one. The observed estimate lies below the mean of the placebo distribution. This does not show that streaming has no effect. It shows that, with these tools and these data, the reported effect cannot be cleanly distinguished from a generic Friday-versus-non-Friday contrast.

4.2 Pre-Treatment Anomalies

A well-identified event-study should produce small, noisy estimates on pre-treatment days. The day-by-day estimates in our replication do not.

The day-0 estimate is $+16.1$ ($t = 3.20$), and day-1 is $+8.8$ ($t = 2.05$). Day -6, however, shows a large positive effect of $+16.5$ ($t = 3.54$), six days before any release. The joint F-test of flatness across the ten pre-treatment days returns $F = 2.04$ ($p = 0.03$).

The day -6 coefficient has a mechanical interpretation. In a sample where nine of ten releases fall on Friday, day -6 of the event window is, for nine of ten albums, the previous Friday. The event-study therefore recovers a large coefficient on the previous Friday because the previous Friday is also a Friday. This reinforces the concern from Section 4.1: the estimator is picking up part of a Friday pattern rather than isolating a release-day effect alone.

4.3 Out-of-Sample Behavior

A causal parameter tied to streaming-induced distraction should, absent a break in the underlying behavior or technology, continue to appear for major releases after the original sample. We therefore assemble a Tier 3 sample of seven major releases from 2023–2024, each with first-day streaming volumes comparable to or exceeding many albums in the original sample.

Applying the paper's specification to Tier 3 yields a pooled estimate of -8.0 deaths ($SE = 7.56$). The unweighted mean of per-album effects is -2.8 deaths. Neither estimate is distinguishable from zero, and both carry the opposite sign of the original estimate.

Table 2: Out-of-Sample Forecasting Results (2023–2024)

Artist	**Album**	**Streams (M)**	**Excess Deaths**
Taylor Swift	*The Tortured Poets Department*	313	−2.1
Travis Scott	*UTOPIA*	128	10.5
Drake	*For All The Dogs*	109	−12.8
Beyoncé	*Cowboy Carter*	76	−0.4
Billie Eilish	*Hit Me Hard and Soft*	73	7.0
SZA	*SOS*	68	9.4
Morgan Wallen	*One Thing at a Time*	52	−1.5
Average Out-of-Sample Effect:			**+1.4**
Pooled Regression Estimate (SE):			**-8.0 (7.0)**

The post-2022 reversal is suggestive, not decisive. The Tier 3 sample is smaller, and the confidence interval straddles zero. Still, the result removes any easy claim that the pattern is a stable causal regularity. At most, it appears to be sample-specific.

4.4 Absent Positive Dose-Response

Across the 20 Tier 1 and Tier 2 albums, the Pearson correlation between first-day streaming volume and per-album effect is -0.17. The sign is opposite to what the proposed mechanism would

naturally imply. A regression of per-album effect on log first-day streams yields a slope of -7.3 ($SE = 9.8$), imprecise but again wrong-signed.

The raw numbers illustrate the pattern. *Midnights* generated 184.6 million first-day streams and is associated with -1.8 deaths. *Her Loss* generated 97.4 million and is associated with $+57.2$ deaths. *Happier Than Ever* generated 60 million and is associated with $+37.7$ deaths. The ordering of fatality effects does not track the ordering of streaming volume.

Two defenses remain possible. First, the mechanism may saturate, so that all major releases lie above the range where additional streams produce additional deaths. Second, first-day streaming counts may be a noisy proxy for the relevant treatment, namely concurrent in-vehicle streaming. Neither defense is directly testable with the available data. Both are therefore ways to preserve the hypothesis, not affirmative evidence for it.

4.5 Outlier Dependence

Leave-one-out analysis shows that the pooled estimate depends heavily on a single observation. Removing *Her Loss* (November 4, 2022) reduces the pooled mean from 23.0 to 18.9, an influence of $+4.1$ deaths. Removing *Midnights* raises the pooled mean to 25.2, an influence of -2.2. The remaining albums have smaller effects on the pooled result.

The arithmetic is transparent. *Her Loss* posts a local effect of $+59.5$, well outside the distribution of the other nine releases. Whatever happened on that date contributes an outsized share of the pooled estimate. This does not invalidate the finding, but it does narrow what one can infer from it. A broad behavioral mechanism should not depend so heavily on one treatment unit.

4.6 Placebo Outcomes

We apply the paper's specification to FARS variables that should not respond to album releases under any plausible streaming mechanism.

Outcome	Effect	t
Mean crash latitude (degrees)	+0.36	3.02
Crashes at railroad crossings (%)	−0.02	−2.08
Crashes in work zones (%)	−0.60	−1.80
Mean vehicles per crash	+0.00	0.04
Mean persons per crash	−0.03	−0.58

On release days, fatal crashes occur on average 0.36 degrees of latitude farther north than on control days, with $t = 3.02$. The share of fatal crashes at railroad crossings is also lower on release days. These outcomes do not plausibly respond to music streaming.

The most natural interpretation is that the design is capturing seasonal and geographic selection onto particular calendar positions. Latitude in fatal crashes varies over the year as the geographic composition of national traffic risk shifts. If the same calendar-position effects move both placebo outcomes and total fatalities, then part of the headline estimate may reflect that broader selection rather than a release-day streaming channel.

5 Limitations of This Critique

This analysis does not definitively resolve the causal question, and the weaknesses of the critique should be stated plainly.

First, the Friday-placebo exercise shows that the paper's specification cannot cleanly separate a release-day effect from a Friday baseline effect within these data. It does not show that streaming causes zero deaths. Because actual album releases and Fridays coincide in this sample, the two cannot be disentangled without a different design.

Second, the post-2022 reversal is based on only seven albums and is not estimated with high precision. It is suggestive evidence against stability, not conclusive evidence of absence.

Third, the dose-response test uses first-day streaming counts as a proxy for the treatment of interest. The relevant exposure is concurrent in-vehicle streaming, which is unobserved. A weak or negative relationship with a noisy proxy is therefore only limited evidence against the mechanism.

Fourth, with only ten treatment units in the original sample and 37 in the broader album panel,

all inference is inherently fragile. Small-sample instability affects both the original paper and this critique.

The evidence supports a narrower claim than the headline one. The paper's finding replicates and survives many standard robustness checks, but it is fragile along dimensions that matter for causal interpretation.

6 Conclusion

The main empirical result in Patel et al. (2026) is not a coding artifact. Using the same basic design, we recover a closely similar estimate, and the underlying anomaly persists across many reasonable specifications. That replicated pattern is the starting point for interpretation.

Replication of a coefficient, however, is not the same thing as validation of a mechanism. The same data also show that the treatment days are heavily concentrated on Fridays, that the event-study loads strongly on the previous Friday, that placebo comparisons using random Fridays generate estimates of similar magnitude, that the pattern does not extend cleanly to major post-2022 releases, and that album-level effects do not rise with streaming counts. Placebo outcomes also move in ways the proposed mechanism cannot readily explain. Taken together, these results do not demonstrate that streaming has no effect. They do show that the available design does not cleanly separate a release-day effect from broader day-of-week and calendar-position effects in a very small sample.

The broader lesson is methodological. In settings with noisy daily outcomes and a small number of salient treated dates, narrow-window event studies can yield striking and newsworthy estimates even when the underlying comparison is not well isolated. That point has appeared in related debates over election days and cannabis-related holidays (Zhang and Aronow, 2016; Palayew and Harper, 2019; Brown et al., 2018). The same caution applies here. Driver distraction from smartphones is a serious and well-established safety problem. Whether major album releases generate a distinct national mortality shock through streaming is, on the evidence currently available, still uncertain.

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